

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

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ABSTRACT. We separate weighted strict Radon duality, strong spectral response, and closed form realizations. All varying Hilbert spaces are transported to a fixed reference space and the metric connection is included in every response derivative. Dissipativity makes the finite-dimensional compressed resolvent strictly accretive in the right half-plane; resolved vectors outside the generator domain are handled by a form-level Schur complement. We prove model-specific pressure-functional rigidity, changing-filtration optional projection convergence, and a category whose typed contractions preserve rates, covariances, cotangents, and memory.

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1. ROUND-SEVENTEEN POSITIVE CLOSURE: STRICT DUALS, COVARIANT FORM RESPONSE, AND OPTIONAL-PROJECTION LIMITS

The three topologies used in this paper are kept distinct. The weighted strict topology has countably additive signed Radon measures as its dual. The strong anisotropic/graph topology carries spectral response distributions.

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Memory is defined on a fixed operator core or, more generally, through a closed coercive form. Parameter derivatives are taken after transport to a fixed reference realization and include the metric connection.

1.1. Weighted strict duality. Let E be a Polish path space, $W \geq 1$ continuous with compact sublevels, and

$$C_W(E) = \{F \in C(E) : F/W \text{ vanishes at weighted infinity}\}.$$

Give $C_W(E)$ the weighted strict topology β_W : a net F_i converges when it is uniformly bounded in $\|F/W\|_\infty$ and converges uniformly on every W -compact set after division by W . Let $\mathcal{M}_W(E)$ be the signed Radon measures with $\int W d|\mu| < \infty$.

Theorem 1.1 (Exact strict dual and invariant annihilator). *The continuous dual of $(C_W(E), \beta_W)$ is $\mathcal{M}_W(E)$ under integration. If Θ_t is a continuous weighted flow, then the annihilator of the closed linear span*

$$\mathcal{N} = \overline{\text{span}}^{\beta_W} \{F - F \circ \Theta_t : F \in C_W(E), t \in \mathbb{Q}_+\}$$

is exactly the space of invariant signed Radon measures in $\mathcal{M}_W(E)$. Constants are not in $\mathcal{N}$.

Proof. A β_W -continuous functional is bounded on one set $\{F : \|F/W\|_\infty \leq 1, \text{supp } F \subset K\}$ for every compact K . Riesz representation gives compatible signed Radon measures on an increasing compact exhaustion. Continuity on the bounded set forces uniform summability of the weighted variations, producing one $\mu \in \mathcal{M}_W(E)$. Conversely, weighted tightness and dominated convergence make integration continuous for β_W . Invariance plainly annihilates every coboundary. Conversely, if a signed Radon functional annihilates the rational-time coboundaries, continuity of the flow extends the identity to all times, hence the measure is invariant. A nonzero constant has nonzero integral under a probability and therefore cannot lie in the closed coboundary span. $\square$

1.2. Transported Hilbert and form bundles. Fix one platform α . Let H_η be the physical L^2 realization of its prepared phase, with inner product $(u, v)_\eta = (G_\eta u, v)_0$ on a fixed vector space H_0 , where G_η is a positive boundedly invertible C^2 family. Put $T_\eta = G_\eta^{1/2} : H_\eta \rightarrow H_0$ and transport the generator and projection:

$$\tilde{L}_\eta = T_\eta L_\eta T_\eta^{-1}, \quad \tilde{P}_\eta = T_\eta P_\eta T_\eta^{-1}.$$

The connection is $\nabla_\eta u = \partial_\eta u + \frac{1}{2} G_\eta^{-1} (\partial_\eta G_\eta) u$ in the untransported coordinates.

Let $V \subset H_0$ be a common dense form domain and $a_\eta : V \times V \rightarrow \mathbb{C}$ a closed sectorial form satisfying

$$\Re a_\eta(u, u) + c \|u\|_{H_0}^2 \geq c_0 \|u\|_V^2,$$

with two bounded parameter derivatives as forms $V \rightarrow V^*$.

Theorem 1.2 (Covariant type-(B) resolvent response). *The transported operators form a Kato type-(B) family. For $\Re z > z_0$,*

$$D_\eta(z - \tilde{L}_\eta)^{-1} = (z - \tilde{L}_\eta)^{-1}(D_\eta \tilde{L}_\eta)(z - \tilde{L}_\eta)^{-1}$$

as an identity $V^ \rightarrow V$, where $D_\eta \tilde{L}_\eta$ includes the derivative of T_η , and hence the metric connection. In untransported coordinates this is the covariant derivative formula; no Hilbert-metric term is omitted.*

Proof. The uniform coercive estimate gives an isomorphism $z + A_\eta : V \rightarrow V^*$ by Lax–Milgram for $\Re z$ sufficiently large. The form derivatives are bounded in $\mathcal{L}(V, V^*)$, so the inverse map is C^2 and differentiation of $(z + A_\eta)R_\eta(z) = I$ yields the formula. Since T_η and T_η^{-1} are C^2 , differentiating the transported generator explicitly produces the commutator with $(\partial_\eta T_\eta)T_\eta^{-1}$, which is exactly the metric-connection term. $\square$

1.3. Accretive compressed resolvents and form-level memory. Let $\mathcal{R} \subset V$ be finite dimensional. If $\mathcal{R} \subset D(\tilde{L}_\eta)$, let P_η be its orthogonal projection and define

$$C_\eta(z) = P_\eta(z - \tilde{L}_\eta)^{-1}P_\eta.$$

If only $\mathcal{R} \subset V$, define the instantaneous compression by the matrix $a_\eta(r_i, r_j)$ and every Schur complement through the weak form.

Theorem 1.3 (Dissipative compression and memory response). *Assume the prepared Markov generator is dissipative on H_0 . Then, for $\Re z > 0$ and $0 \neq r \in \mathcal{R}$,*

$$\Re(C_\eta(z)r, r)_{H_0} = \Re z \|(z - \tilde{L}_\eta)^{-1}r\|^2 - \Re(\tilde{L}_\eta y, y) > 0,$$

where $y = (z - \tilde{L}_\eta)^{-1}r$. Thus $C_\eta(z)$ is invertible in the right half-plane. The memory transform

$$\widehat{K}_\eta(z) = zP_\eta - P_\eta \tilde{L}_\eta P_\eta - C_\eta(z)^{-1}$$

is well typed on a common operator core; on a mere form domain the same formula holds with $P\tilde{L}P$ replaced by the form matrix. Its covariant derivative consists of the derivatives of C , P , the form, and the metric connection, all on the fixed reference realization.

Proof. The displayed identity follows from $r = (z - \tilde{L})y$ and dissipativity. Strict positivity gives injectivity and finite dimensionality gives invertibility. The common operator-core formula is then algebraic. In the form setting, block the weak equation $(z + A)y = f$ with respect to $\mathcal{R} \oplus \mathcal{R}^\perp$ and eliminate the second component; the resulting finite matrix is the form-level Schur complement. Differentiate after the fixed transport of Theorem 1.2; the inverse derivative is $D(C^{-1}) = -C^{-1}(DC)C^{-1}$ and every projection/metric term remains visible. $\square$

1.4. Model-specific rigidity. For a potential F on one platform, let

$$\mathcal{P}_F(G) = Q(F + G) - Q(F)$$

be the full perturbed pressure functional on a neighborhood of zero in the model's strong source space.

Theorem 1.4 (Pressure-functional rigidity). *If two regular potentials F_1, F_2 on the same platform satisfy*

$$\mathcal{P}_{F_1}(G) = \mathcal{P}_{F_2}(G)$$

for every G in one strong neighborhood, then their equilibrium path laws coincide. For the Sinai platform their difference is a constant plus a closed dynamical coboundary in the A3 potential space. For the hard-sphere platform their dual fields differ by the closed balance gauge $(r, -\Delta r)$ and the finite collision invariants.

Proof. Differentiating in every strong source direction gives equality of all local path-cylinder expectations. The cylinder algebra is convergence determining and dense in the strict topology on weighted sublevels, hence the equilibrium Radon laws coincide. On the Sinai platform, periodic orbit approximation for the proved countable renewal code implies that all periodic sums of $F_1 - F_2$ equal one constant times the period. The Livšic theorem on the A3 summable-variation/weighted space and closedness of the coboundary range give the representation. On the hard-sphere platform, equality against all balanced finite-action variations and the closed-range theorem of B3 identify the annihilator as the full balance gauge plus invariants. The argument uses the complete perturbation functional, not one scalar pressure value. $\square$

1.5. Changing filtrations and optional projections. Let $(X^\varepsilon, L^\varepsilon)$ be the microscopic path and its canonical likelihood ratio, and let $\mathcal{G}_t^\varepsilon$ be the coarse observation filtration. Assume the A4/C1 conditional kernels converge in the regular chart and the likelihood ratios are uniformly bounded in $L^{1+\delta}$.

Theorem 1.5 (Extended weak convergence of optional projections). *The triples*

$$(X^\varepsilon, L^\varepsilon, \mathbf{E}[L_T^\varepsilon \mid \mathcal{G}_t^\varepsilon])$$

converge in the extended weak topology to the corresponding limiting path, likelihood, and optional projection. Consequently the projected likelihood is the limiting stochastic exponential, and the Girsanov and bounded risk-sensitive BSDE representations commute with the microscopic limit.

Proof. Convergence of the conditional kernels means that, for every bounded continuous future cylinder F , the conditional expectation $E[F(X^\varepsilon) \mid \mathcal{G}_t^\varepsilon]$ converges jointly with the past to its limiting kernel evaluation, uniformly on compact regular histories. Approximate L_T in L^1 by bounded continuous cylinders using the uniform $L^{1+\delta}$ bound. The approximation and kernel convergence identify all finite-dimensional optional projections; Doob's maximal

inequality gives tightness. The limiting Gaussian likelihood from B3/A4 has the Cameron–Martin stochastic-exponential form, and standard Girsanov and exponential-transform identities then yield the representations. $\square$

1.6. The typed contraction category. Objects are tuples $(E, \tau, I, \mathcal{X}, H, V, L)$ consisting of a platform label, path topology, rate, strong source space, Hilbert realization, form domain, and generator. A morphism consists of a continuous or exponentially approximable state map, its strict adjoint on Radon tests, a bounded pullback on strong sources, and an intertwining map on the transported form bundle.

Theorem 1.6 (Naturality of rates, tangents, and memory). *For every typed morphism T , contraction of rates and pullback of pressure commute,*

$$I_T(y) = \inf_{Tx=y} I(x), \quad Q_T(G) = Q(T^*G).$$

At regular exposed points, Gaussian covariance and cotangents push forward as $T\Sigma T^$ and T^* , and compressed memory commutes whenever the resolved subspace and form intertwiner form a commuting square. Composition of morphisms preserves these identities.*

Proof. The pressure identity is exact at finite volume and the rate identity is the extended contraction theorem. Differentiating the exact pressure identity on the transported strong spaces gives the covariance and cotangent formulas. The form-level Schur complement is natural under a commuting block intertwiner, giving memory compatibility. All maps and topologies are part of the object/morphism definition, so composition follows by ordinary composition rather than by a bookkeeping declaration. $\square$

Theorem 1.7 (Round-seventeen C2 closure). *Each platform has a correct signed-Radon strict dual, a transported strong response realization with metric connection, a well-typed accretive form/operator memory, model-specific pressure rigidity, convergent optional projections, and a genuine typed contraction functor.*

Proof. Combine Theorems 1.1, 1.2, 1.3, 1.4, 1.5, and 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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